

RESUME

Rupesh Kumar

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Objective:

To pursue an opportunity involving active research, design and development in IT Field and eventually create an unchallengeable image of myself in the IT industry

Academic Record:

Degree Offered	Board / University	Year of Passing	CPI / Percentage Obtained
M.Tech (CSE)	IIT, Guwahati	May, 2008	8.84 (upto II nd semester)
MCA	IGNOU (MSIT, New Delhi)	Dec, 2006	65.67
B.Sc (Comp.Science)	Delhi University	May, 2004	64.66
BIT	IGNOU (PCTI, New Delhi)	Sep, 2003	64.68
10+2	C.B.S.E (New Delhi)	2000	67.40
SSC	C.B.S.E (New Delhi)	1998	61.60

Accolades:

- AIR 473 in GATE 2006 in CS
- 91.21 Percentile in GATE 2005
- Selected for the SET-DRDO 2005 interview
- Participated in Science Fair in District level in 10+2

Courses taken:

- **Sem-I** : Compilers, DBMS, CN, OS, Wireless sensor network, Computational Number theory and Cryptography.
- **Sem-II**: Linux Kernel Prog., Wireless Networks, Network Security, Hierarchical memory algorithms.

Technical Skill set:

- **Have in-depth understanding on Data Structures and Algorithms**
- **Languages** : C, C++, Java, Shell Programming
- **Operating Systems** : Windows XP, Linux
- **Databases** : Oracle, MS-SQL 7.0, MS-Access
- **Tools** : LEX, YACC, NS-2.29

Experience:

- Fresher

Academic Project Details:

1. M.Tech Project : ***Optimal Greedy Routing in Ad-Hoc Network***

Guide : Dr. S.V.Rao (Asst. Prof. of CSE Dept., IIT, Guwahati)

Language : ns 2.29

Project Description:

Recent availability of small inexpensive low power GPS receivers and techniques for finding relative coordinates based on signal strengths, and the need for the design of power efficient and scalable networks, provided justification for applying position based routing methods in ad hoc networks. We proposed a **Optimal Greedy Routing protocol**, in which in beaconing interval instead of just sending its position like in GPSR it also sends speed and direction of movement, so using this direction information a neighboring node can determine the exact position of the node at time t_1 . So forwarding node can decide more efficiently next forwarding node. As compared to GPSR and ARP it selects the next forwarding node which is more optimal. Implementation of the protocol will be done in the **ns2.29**, and its result will be checked in high mobility scenario with varying beaconing interval (pause time).

2. MCA Project : ***K-Map Simplification***

Guide : Mr. Munesh Chuahan (Asst. Prof. of CSE Dept, GEIT Deharadun)

Language : C language

Project Description:

To reduce the number of terms or literals in Boolean expression representing a circuit, there is graphical method, called a **Karnaugh map or K-map**. K-map provides us a visual method for simplifying sum-of-products expansions. Unfortunately, minimizing Boolean functions with many variables is a computationally intensive problem. It has been shown that this problem is an NP-complete problem means there exist no polynomial time algorithm for minimizing Boolean circuits unlikely.

3. B.Sc(Comp. Science)

Project : ***Automation of Admission process in Shivaji college of DU***

Guide : Mrs. Preeti Sharma (Asst. Prof. & HOD of CS Dept., Shivaji college Delhi University)

Language : Front-end: Visual Basic

Back-end: Ms-Access

Project Description:

Due to multiple window and manual admission process there is so much problem is faced by the students, and office staff is also not able to maintain the such large database of the students. So there is need to automate the admission process in the college which leads to the convenience to the students. Delay in the admission process can be avoided.

4. BIT Project : ***Implementation of Deffie Hellman Key exchange with RSA based signature***

Guide : Mr. Akshay gupta (faculty at PCTI New Delhi)

Language : Java

Project Description:

If someone know about the Deffie Hellman and RSA, then this project is self explanatory. Anyway in this project we used the RSA to exchange the Deffie Hellman key because as we know the Deffie Hellman suffers from the Man-In-The-Middle attack. That is using RSA private key we encrypt the Deffie Hellman key and other side will decrypt the key.

5. M.Tech(II Sem)

Term end Project : *MSPIN & DEAQP protocol in Sensor Networks*

Guide : Dr. S.Nandi (Prof. of CSE Dept., IIT, Guwahati)

Language : ns2.29

Project Description:

I have given two new protocols based on Data Centric routing and Hierarchical routing protocols for sensor networks named **Modified SPIN (MSPIN)** and **Dynamic Energy Aware Query Processing(DEAQP)** in sensor networks. Data-centric technologies are needed that perform in-network aggregation of data to yield energy-efficient dissemination of data. Moreover, I have minimized the energy requirement for dissemination of data in MSPIN and the query processing in the sensor networks in DEAQP protocol.

Other Academic Projects:

- Implementation of syntax analyzer for select statement of database using LEX and YACC Tools
- Developed Project on Hotel Management System in VB
- Implemented Online Library System for CSE Department (Involved in Design only)

Seminar & Workshop attended:

- Ethical Hacking in IIT, Guwahati.
- ICDCN-06 in IIT, Guwahati.
- RE-07 15th IEEE organized by IIT, Delhi and IIT, Kanpur.
- Preparedness of Disaster Management in IIT Guwahati.

References:

1. Dr. S.V.Rao, Assistant Professor, Department Of Computer Science & Engineering, IIT, Guwahati, svrao@iitg.ernet.in Ph: 0361 2582353, Mobile: 09954191667
2. Dr. S.Nandi, Professor, Department Of Computer Science & Engineering, IIT, Guwahati, sukumar@iitg.ernet.in Ph: +91 361 2582357

Strengths:

- Hard working,
- Innovative,
- Committed to Learn and Lead,
- Driven by spirits of service and ability to work in crisis.

Personal Details:

- **Date of Birth** : 15 July 1983
- **Gender** : Male
- **Hobbies** : Yoga, Meditation, Reading Books, Listening music, watching movies, Swimming, Badminton.
- **Languages** : Hindi, English